

Quadratic functions

Find the turning point. Check the sign. Use the model.

01 Read the quadratic

$$y = a(x - h)^2 + k$$

Vertex (h, k) • axis of symmetry $x = h$

$$a > 0$$

Minimum value = k

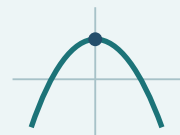
Range: $y \geq k$



$$a < 0$$

Maximum value = k

Range: $y \leq k$



02 Complete the square to find the turning point

First factor out the coefficient of x^2 from the x terms.

$$f(x) = 2x^2 - 8x + 5$$

$$= 2(x^2 - 4x) + 5$$

$$= 2[(x - 2)^2 - 4] + 5$$

$$= 2(x - 2)^2 - 3$$

READ OFF

$$x = 2$$

$$\text{min} = -3$$

$$y \geq -3$$

Why it works: $(x - 2)^2$ is never negative, so the smallest value is at $x = 2$.

03 Always positive or always negative?

For all real x , use the lowest or highest point of $y = a(x - h)^2 + k$.

ALWAYS POSITIVE

$$a > 0 \text{ and } k > 0$$

ALWAYS NEGATIVE

$$a < 0 \text{ and } k < 0$$

If $k = 0$, the graph touches $y = 0$. It is not strictly positive or negative.

04 Use it as a model

A rectangle has sides x cm and $(12 - x)$ cm. What is its greatest area?

$$A(x) = x(12 - x) = -(x - 6)^2 + 36$$

Domain: $0 < x < 12$. Maximum area: 36 cm^2 when $x = 6$ cm.

In word problems, check the allowed x -values before using the vertex.

EXAM CHECK

Show the square • State both x and the max/min • Include units and domain